

DERIK COLUCCI

Westlake, FL | 561-779-8558 | Derik.Colucci13@gmail.com | LinkedIn | GitHub

Software Development & Programming | Machine Learning | Algorithm Development

I am a motivated and detail-oriented Computer Engineering student with proven expertise in software development, machine learning, and algorithm design. Experienced in building scalable automated data pipelines, web scraping frameworks, and full-stack applications using Python, C++, React, and C#. I am skilled in deep learning model development with TensorFlow and PyTorch, data analysis, and predictive modeling. Strong foundation in problem-solving, technical communication, and cross-functional collaboration, with a demonstrated ability to deliver impactful technology solutions in dynamic environments. Actively pursuing internship opportunities to contribute advanced programming skills and innovative machine learning techniques to real-world projects.

CORE COMPETENCIES

Data Structures and Algorithms | Technical Documentation & Reporting | Data Analysis & Problem Solving | Data Pipeline Development & Integration | Neural Network Design & Optimization | Reinforcement Learning | Dynamic Web Scraping & Parsing | Software Development | Simulation & Predictive Modeling | Object Oriented Programming | API Development

EDUCATION

B.S. in Computer Engineering (GPA: 3.84)

Florida Atlantic University | Expected Date: 2026

B.S in Exercise Physiology

Florida State University | 2019

TECHNICAL SKILLS

- Dense Neural Network (TensorFlow):** Python, C++, C#, React, SQL
- Machine Learning & Deep Learning:** TensorFlow, PyTorch, Keras, OpenCV, Neural Networks (Dense, CNNs), Reinforcement Learning, Model Optimization
- Data Science & Analytics:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Data Visualization, Predictive Analytics, Algorithm Development
- Web Development & Automation:** API Development, Selenium WebDriver, BeautifulSoup, Web Scraping, Front-End UI/UX Design, Email Automation (SMTP)
- Data Engineering & Pipeline Development:** Data Pipeline Automation, Data Integration, Regex, OS Automation, Database Design & Management (PostgreSQL, Supabase), SQL Querying
- Software Tools & Technology:** Git, Jupyter Notebook, Google Colab, VS Code, MATLAB
- Other Skills:** Neural Networks, CNNs, Data Visualization, CAD, Pipeline Development for Data Collection & Analysis, Web Scraping Techniques, APIs

EXPERIENCE

PairTech - Software Development and Data Science Intern
Boca, FL

May 2025 – Oct 2025

- Engineered an Automated Web Scraping Pipeline:** Developed a Python-based system using Selenium and BeautifulSoup to automate extraction of company names and industry data from dynamic websites via a headless browser.
- Entity Extraction & Industry Classification:** Created a multi-tiered algorithm to identify company names from metadata and HTML elements, then classified them into industries and sub-industries using a keyword-based NLP system.

- **Scheduling System Development:** Built a full-featured scheduling system for PairTech’s website using React for the front end and C# for the back end..
- **Optimized System Performance with Monitoring:** Integrated comprehensive timing and benchmarking tools to measure and enhance the efficiency of parsing and categorization workflows.
- **Front-End UI Improvements:** Designed responsive, modular components using React Hooks and best practices for improved user interaction and maintainability.

PERSONAL PROJECTS

Neural Network Portfolio Jan 2025 – May 2025
Deep Learning & Computer Vision | TensorFlow, PyTorch, Python

- **Dense Neural Network (TensorFlow):** Engineered and trained a fully connected deep learning model to accurately classify handwritten digits from the MNIST dataset, achieving an impressive 96.1% test accuracy by fine-tuning hyperparameters and utilizing advanced optimization techniques.
- **Convolutional Neural Network (TensorFlow):** Developed a high-performance Convolutional Neural Network for digit recognition using TensorFlow, achieving a 98.5% test accuracy by leveraging convolutional layers, pooling, and dropout techniques to effectively capture spatial features and prevent overfitting.
- **Convolutional Neural Network (PyTorch):** Implemented a robust Convolutional Neural Network in PyTorch for MNIST digit classification, achieving a 98.3% test accuracy, utilizing data augmentation, batch normalization, and advanced training strategies to enhance model generalization and improve performance.

Crypto Correlation Analysis Dec 2024 – Jan 2025
Predictive Analytics & Algorithm Development | Python, Pandas, Matplotlib

- **Cross-Asset Volatility:** Engineered predictive analytics solution to assess cryptocurrency market dynamics, focusing on Dogecoin’s volatility as influenced by Bitcoin’s trading patterns and historical performance trends.
- **Algorithm Development:** Developed decision-making algorithm by simulating real-world market scenarios to optimize buy/sell strategies, leveraging Bitcoin-driven market triggers to maximize cryptocurrency investments.
- **Python Programming:** Implemented Python-based analytics pipelines while utilizing libraries like Pandas and Matplotlib to transform raw market data into comprehensive visual insights for strategic forecasting.
- **Risk Reduction Strategies:** Introduced innovative data-driven methodologies, evaluating cross-asset price behaviors and delivering impactful insights to inform portfolio management and reduce market risks.

Python Keylogger with Email Capability Dec 2024
Software Development & Automation | Python, SMTP

- **Keylogger Development:** Designed and implemented highly efficient keylogger, accurately recording user keystrokes such as special keys, with real-time backspace correction for data integrity.
- **Automated Reporting:** Established keyword detection functionality using advanced pattern-matching algorithms, automating actionable insights and periodic keyword usage summaries sent via email.
- **SMTP Protocols:** Built secure email notification system, leveraging SMTP protocols to deliver encrypted alerts triggered by dynamic log size thresholds for timely reporting.
- **Data Management:** Streamlined data handling workflows by implementing automated log rotation and cleanup processes to optimize performance and prevent redundancy in data storage.

CAD Design & Prototyping | CAD Software, 3D Printing

- **CAD Design Expertise:** Engineered unique guitar pick design using CAD software by integrating string-clip feature for convenient storage without compromising playability or sound quality.
- **Prototyping & Testing:** Executed prototype development through advanced 3D printing techniques, confirming precise functionality and durability before transitioning to small-scale manufacturing.
- **Collaboration & Cost-Effective Scaling:** Cooperated with third-party manufacturers to refine production processes, achieving cost-effective scalability for market-ready product delivery.
- **Patent Application:** Independently authored and filed provisional patent application with the U.S. Patent Office aiming at safeguarding intellectual property and establishing product exclusivity.

COLLABORATIVE PROJECTS

BuffBuds – Full Stack Fitness App

Aug 2025-Present

Backend & Full-Stack Development | JavaScript, React, Supabase, PostgreSQL

- **Backend API Development:** Designed and implemented backend logic to support workout tracking, user authentication, and data validation using Supabase and custom RESTful endpoints.
- **Database Architecture:** Built and optimized a scalable PostgreSQL schema on Supabase to manage users, workouts, exercises, and training programs.
- **Cross-Functional Collaboration:** Worked in an agile team environment, participating in sprint planning, code reviews, and feature development across the stack.
- **Security & Authentication:** Implemented secure login and session handling using Supabase auth, with role-based access control and protected routes.

Exoskeleton Suit – Baptist Health Sponsored Project

Aug 2025-Present

Biomechanical Engineering & Machine Learning | Python, Embedded Systems, TensorFlow

- **ML-Based Movement Prediction:** Partnered with Baptist Health and the chief orthopedic surgeon to co-develop a wearable exoskeleton for mobility support and rehabilitation.
- **Clinical Collaboration:** Built and optimized a scalable PostgreSQL schema on Supabase to manage users, workouts, exercises, and training programs.
- **Hardware Integration:** Coordinated software–hardware interfacing to translate motion predictions into mechanical actuation using microcontrollers and custom-built motor control systems.
- **Iterative Prototyping:** Led data collection, model training, and continuous refinement cycles to improve response time, accuracy, and adaptability of suit control logic.